Random Graphs

Lecture Notes

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These notes develop the lecture into a self-contained introduction to the probabilistic method in graph theory. The notation follows the treatment of random graphs in Diestel’s *Graph Theory*. Arguments abbreviated during the lecture are supplied in full.

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1 Graphs and notation

Throughout the notes, graphs are simple and undirected. They are finite except in Section 6, where countably infinite graphs are considered.

1.1 Graphs, isomorphisms, and subgraphs

Definition 1.1. A *graph* is a pair $G = (V, E)$, where V is a set of vertices and

$$E \subseteq [V]^2 := \{\{x, y\} : x, y \in V,\ x \neq y\}.$$

The *order* and *size* of G are denoted, in Diestel's notation, by

$$|G| := |V|, \qquad \|G\| := |E|.$$

The quantity

$$\varepsilon(G) := \frac{\|G\|}{|G|}$$

is called the *density* of G. Thus the average degree of G is $2\varepsilon(G)$.

An edge $\{x, y\}$ is usually written simply as xy. The complete graph on k vertices is denoted by K_k.

Definition 1.2. Two graphs $G = (V, E)$ and $G' = (V', E')$ are *isomorphic*, written $G \simeq G'$, if there is a bijection $\varphi : V \to V'$ such that

$$xy \in E \quad \Longleftrightarrow \quad \varphi(x)\varphi(y) \in E' \qquad (x, y \in V).$$

Such a map φ is an *isomorphism*.

Definition 1.3. Let $G = (V, E)$ and $G' = (V', E')$. We call G' a *subgraph* of G, and write $G' \subseteq G$, if $V' \subseteq V$ and $E' \subseteq E$.

For $U \subseteq V$, the *induced subgraph* of G on U is

$$G[U] := (U, E \cap [U]^2).$$

Thus an ordinary subgraph may omit edges between its chosen vertices, whereas an induced subgraph must retain every such edge of the original graph.

1.2 Standing notation

Notation	Meaning
$[n] = \{1, \dots, n\}$	the standard n-vertex set
$[V]^2$	the set of unordered pairs of distinct elements of V
$q = 1 - p$	the probability that a potential edge is absent
$(n)_k = n!/(n-k)!$	the falling factorial
C_k	the cycle of length k
$p \ll t$, $p \gg t$	$p/t \to 0$, respectively $p/t \to \infty$

We repeatedly use the union bound

$$\mathbb{P}\left(\bigcup_i A_i\right) \le \sum_i \mathbb{P}(A_i)$$

and the elementary inequality $1 - x \le e^{-x}$ for $0 \le x \le 1$.

2 The binomial random graph

2.1 The probability space $\mathcal{G}(n,p)$

Fix the vertex set $V = [n]$ and a number $p \in [0,1]$. Put $q = 1 - p$.

Definition 2.1. The *binomial random graph* $\mathcal{G}(n,p)$ is the probability space in which every possible edge $e \in [V]^2$ is included independently with probability p.

Formally, for each $e \in [V]^2$ take the two-point probability space

$$\Omega_e = \{0_e, 1_e\}, \qquad \mathbb{P}_e(1_e) = p, \quad \mathbb{P}_e(0_e) = q,$$

and form the product space

$$\Omega = \prod_{e \in [V]^2} \Omega_e.$$

For $\omega \in \Omega$, the corresponding graph has edge set $E(\omega) = \{e : \omega(e) = 1_e\}$. If A_e is the event that e is present, then the events $(A_e)_{e \in [V]^2}$ are mutually independent and $\mathbb{P}(A_e) = p$. Consequently, any particular graph F on $[n]$ has

$$\mathbb{P}(G = F) = p^{\|F\|} q^{\binom{n}{2} - \|F\|}.$$

The notation $G \in \mathcal{G}(n,p)$ means that G is sampled from this distribution. The parameter p may depend on n, even when this is not displayed.

2.2 Fixed subgraphs and induced copies

Let H be a fixed graph on a prescribed set $U \subseteq [n]$, with $|H| = k$ and $\|H\| = \ell$. Only the ℓ edges of H are required for $H \subseteq G$, so

$$\mathbb{P}(H \subseteq G) = p^{\ell}.$$

For H to appear as the induced graph $G[U]$, all other possible edges in $[U]^2$ must be absent. Hence

$$\mathbb{P}(G[U] = H) = p^{\ell} q^{\binom{k}{2} - \ell}.$$

This distinction—requiring certain edges versus prescribing all edges and non-edges—will recur throughout the notes.

3 Ramsey numbers by the probabilistic method

3.1 Independent sets and cliques

Definition 3.1. An *independent set* in G is a set of pairwise nonadjacent vertices. The *independence number* $\alpha(G)$ is the maximum size of an independent set.

A *clique* is a set of pairwise adjacent vertices. The *clique number* $\omega(G)$ is the maximum size of a clique.

For a fixed k-set $S \subseteq [n]$, the probability that S is independent is $q^{\binom{k}{2}}$, while the probability that S is a clique is $p^{\binom{k}{2}}$. The union bound therefore gives the following estimates.

Proposition 3.2. *For $n \geq k \geq 2$ and $G \in \mathcal{G}(n,p)$,*

$$\mathbb{P}(\alpha(G) \geq k) \leq \binom{n}{k} q^{\binom{k}{2}}, \qquad \mathbb{P}(\omega(G) \geq k) \leq \binom{n}{k} p^{\binom{k}{2}}.$$

The events corresponding to different k-sets need not be independent; the union bound does not require independence.

3.2 The diagonal Ramsey lower bound

Definition 3.3. The diagonal Ramsey number $R(k) = R(k,k)$ is the least integer N such that every graph on N vertices contains either a K_k or an independent set of size k.

Theorem 3.4 (Erdős, 1947)**.** *For every integer $k \geq 4$,*

$$R(k) > 2^{k/2}.$$

Proof. Set $p = q = 1/2$ and $n = \lfloor 2^{k/2} \rfloor$. Since $k! > 2^k$ for $k \geq 4$,

$$\begin{aligned} \binom{n}{k} 2^{-\binom{k}{2}} &< \frac{n^k}{2^k}\, 2^{-k(k-1)/2} \\ &\leq 2^{k^2/2 - k - k(k-1)/2} = 2^{-k/2} < \frac{1}{2}. \end{aligned}$$

By Proposition 3.2,

$$\mathbb{P}(\alpha(G) \geq k \text{ or } \omega(G) \geq k) \leq 2\binom{n}{k} 2^{-\binom{k}{2}} < 1.$$

Thus some graph on n vertices has neither a k-clique nor an independent k-set. Hence $R(k) > n$; since $R(k)$ is an integer, this implies $R(k) > 2^{k/2}$. □

The proof does not construct the desired graph. It proves its existence by showing that a random graph fails the unwanted conditions with positive probability. This is the basic form of the *probabilistic method*.

4 First-moment estimates

4.1 Random variables and indicator variables

Definition 4.1. A random variable on $\mathcal{G}(n,p)$ is a function $X : \mathcal{G}(n,p) \to \mathbb{R}$. Its expected value is

$$\mathbb{E}X = \sum_G \mathbb{P}(G)X(G),$$

where the sum ranges over all graphs on $[n]$.

Expectation is linear:

$$\mathbb{E}\left(\sum_i X_i\right) = \sum_i \mathbb{E}X_i,$$

whether or not the random variables X_i are independent. For an event A, its indicator variable is $\mathbf{1}_A = 1$ on A and 0 otherwise; then $\mathbb{E}\mathbf{1}_A = \mathbb{P}(A)$.

Lemma 4.2. *For $k \geq 3$, the expected number of k-cycles in $G \in \mathcal{G}(n,p)$ is*

$$\frac{(n)_k}{2k}p^k.$$

Proof. There are $(n)_k$ ordered sequences of k distinct vertices. Every unoriented k-cycle is represented $2k$ times: there are k choices of starting vertex and two directions. Hence the complete graph on $[n]$ contains $(n)_k/(2k)$ distinct k-cycles.

For each such cycle C, let $X_C = \mathbf{1}_{\{C \subseteq G\}}$. The cycle has k edges, so $\mathbb{E}X_C = p^k$. If $X = \sum_C X_C$, linearity gives

$$\mathbb{E}X = \sum_C \mathbb{E}X_C = \frac{(n)_k}{2k}p^k.$$

□

Lemma 4.3 (Markov's inequality)**.** *If $X \geq 0$ and $a > 0$, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}.$$

Proof. Pointwise, $X \geq a\mathbf{1}_{\{X \geq a\}}$. Taking expectations yields the claim. □

Markov's inequality converts a small expected count into a small probability that the count is large. The next lemma applies this philosophy indirectly to large independent sets.

4.2 A useful bound on the independence number

Lemma 4.4. *Fix $k \in \mathbb{N}$. Suppose that*

$$p = p(n) \geq \frac{16k^2}{n}$$

for all sufficiently large n. If $G \in \mathcal{G}(n,p)$, then

$$\mathbb{P}\left(\alpha(G) \geq \frac{n}{2k}\right) \longrightarrow 0.$$

Proof. For an integer r with $2 \le r \le n$, Proposition 3.2 and $1 - p \le e^{-p}$ give

$$\mathbb{P}(\alpha(G) \ge r) \le \binom{n}{r}(1-p)^{\binom{r}{2}} \le 2^n e^{-p\binom{r}{2}}.$$

Since $\binom{r}{2} \ge r^2/4$ for $r \ge 2$, taking $r = \lceil n/(2k) \rceil$ gives, for large n,

$$\begin{aligned}\mathbb{P}(\alpha(G) \ge r) &\le 2^n e^{-pr^2/4} \\ &\le 2^n \exp\left(-\frac{pn^2}{16k^2}\right) \\ &\le 2^n e^{-n} \longrightarrow 0.\end{aligned}$$

□

5 Large girth and large chromatic number

Definition 5.1. The *girth* $g(G)$ is the length of a shortest cycle in G; by convention $g(G) = \infty$ when G is acyclic. A shortest cycle has no chord, so it is automatically induced.

Definition 5.2. A proper vertex coloring assigns colors to the vertices so that adjacent vertices receive different colors. The *chromatic number* $\chi(G)$ is the minimum number of colors in a proper coloring.

Every color class is independent. Therefore

$$\chi(G) \ge \frac{|G|}{\alpha(G)}. \tag{1}$$

Theorem 5.3 (Erdős, 1959)**.** *For every $k \in \mathbb{N}$, there exists a finite graph H such that*

$$g(H) > k \qquad \text{and} \qquad \chi(H) > k.$$

Proof. It is enough to consider $k \ge 3$. Choose $0 < \varepsilon < 1/k$ and set

$$p = n^{\varepsilon - 1}.$$

Let $G \in \mathcal{G}(n,p)$, and let X be the number of cycles in G whose lengths lie between 3 and k. By Lemma 4.2, and using $np = n^{\varepsilon} \ge 1$,

$$\begin{aligned}\mathbb{E}X &= \sum_{\ell=3}^{k} \frac{(n)_\ell}{2\ell} p^\ell \\ &\le \frac{1}{2}\sum_{\ell=3}^{k} (np)^\ell \\ &\le \frac{k-2}{2}(np)^k = \frac{k-2}{2} n^{k\varepsilon}.\end{aligned}$$

Markov's inequality now yields

$$\mathbb{P}(X \ge n/2) \le \frac{2\mathbb{E}X}{n} \le (k-2)n^{k\varepsilon - 1} \longrightarrow 0,$$

because $k\varepsilon < 1$.

Also $p = n^{\varepsilon-1} \geq 16k^2/n$ for all sufficiently large n. By Lemma 4.4,

$$\mathbb{P}\left(\alpha(G) \geq \frac{n}{2k}\right) \longrightarrow 0.$$

Consequently, for large enough n, there exists a graph G satisfying simultaneously

$$X(G) < \frac{n}{2}, \qquad \alpha(G) < \frac{n}{2k}.$$

Delete one vertex from every cycle of length at most k, choosing at most one new vertex for each of the $X(G)$ cycles, and let H be the resulting induced subgraph. Every short cycle is destroyed, while

$$|H| \geq n - X(G) > \frac{n}{2}, \qquad \alpha(H) \leq \alpha(G) < \frac{n}{2k}.$$

Thus $g(H) > k$, and by (1),

$$\chi(H) \geq \frac{|H|}{\alpha(H)} > k.$$

□

The proof illustrates a standard alteration argument: first obtain a random object with few local defects, and then delete a small number of vertices to remove every defect.

Corollary 5.4. *There are finite graphs with arbitrarily large girth and arbitrarily large minimum degree; hence their average degree can also be arbitrarily large. There are also finite graphs with arbitrarily large girth and arbitrarily large vertex-connectivity.*

Proof. Choose a graph of sufficiently large girth and chromatic number using Theorem 5.3, and then take a vertex-minimal subgraph F with the same chromatic number. If a vertex v of F had $d_F(v) \leq \chi(F)-2$, a $(\chi(F)-1)$-coloring of $F-v$ could be extended to v, a contradiction. Hence

$$\delta(F) \geq \chi(F) - 1.$$

Passing to a subgraph cannot create a shorter cycle, which proves the first assertion.

For the connectivity assertion, combine the first assertion with Mader's standard deterministic theorem that, for every r, sufficiently large average degree forces an r-connected subgraph. Again, taking a subgraph cannot decrease the girth. □

6 Almost-sure properties and the Rado graph

6.1 Graph properties and asymptotic probability

Definition 6.1. A *graph property* $\mathcal{P}$ is a class of graphs closed under isomorphism. We say that *almost every* graph in $\mathcal{G}(n,p)$ has $\mathcal{P}$, or that $\mathcal{P}$ holds *asymptotically almost surely* (a.a.s.), if

$$\mathbb{P}(G \in \mathcal{P}) \longrightarrow 1 \qquad (n \to \infty).$$

Proposition 6.2. *Fix $p \in (0,1)$ and a finite graph H. Almost every $G \in \mathcal{G}(n,p)$ contains an induced copy of H.*

Proof. Write $|H| = k$ and $\|H\| = \ell$, and partition as many vertices of $[n]$ as possible into $m = \lfloor n/k \rfloor$ disjoint k-sets. On each block fix a prescribed labeling of H. The probability that this particular labeled induced copy occurs is the positive constant

$$r = p^{\ell} q^{\binom{k}{2}-\ell}.$$

The m events are independent because they involve disjoint sets of edges. Therefore

$$\mathbb{P}(\text{none of these blocks induces } H) = (1-r)^m \longrightarrow 0.$$

□

6.2 Finite extension properties

Definition 6.3. For $i, j \in \mathbb{N} \cup \{0\}$, let $\mathcal{P}_{i,j}$ be the property that for every two disjoint vertex sets U, W with $|U| \le i$ and $|W| \le j$, there exists a vertex $v \notin U \cup W$ adjacent to every vertex of U and to no vertex of W.

Proposition 6.4. *For every fixed $p \in (0,1)$ and fixed i, j, almost every $G \in \mathcal{G}(n,p)$ has $\mathcal{P}_{i,j}$.*

Proof. Fix disjoint $U, W \subseteq [n]$, with $|U| = u \le i$ and $|W| = w \le j$. A particular vertex $v \notin U \cup W$ has the required adjacency pattern with probability $p^u q^w$. These events are independent as v varies outside $U \cup W$. Thus

$$\mathbb{P}(\text{no suitable } v) = (1 - p^u q^w)^{n-u-w} \le (1 - p^i q^j)^{n-i-j}.$$

There are at most $C_{i,j} n^{i+j}$ possible ordered pairs (U, W), for a constant $C_{i,j}$. The union bound gives

$$\mathbb{P}(G \notin \mathcal{P}_{i,j}) \le C_{i,j} n^{i+j} (1 - p^i q^j)^{n-i-j} \longrightarrow 0,$$

because an exponential decay dominates every fixed polynomial. □

Define the full *extension property* by

$$\mathcal{P} := \bigcap_{i,j \ge 0} \mathcal{P}_{i,j}.$$

For a countable graph, this says that every prescribed finite pattern of adjacencies and non-adjacencies is realized by some new vertex.

6.3 Existence and uniqueness of the Rado graph

Theorem 6.5 (Rado graph)**.** *Up to isomorphism, there is a unique countably infinite graph R with the extension property $\mathcal{P}$. It is called the* Rado graph.

Existence. Start with a one-vertex graph R_0. Having constructed the finite graph R_m, form R_{m+1} as follows: for every subset $U \subseteq V(R_m)$, add a new vertex v_U adjacent to precisely the vertices of U among the old vertices. For definiteness, put no edges between vertices added at the same stage. Let

$$R = \bigcup_{m \ge 0} R_m.$$

This graph is countable. If $U, W \subseteq V(R)$ are finite and disjoint, choose m so that $U \cup W \subseteq V(R_m)$. At the next stage the vertex v_U is adjacent to every vertex in U and none in W. Hence R has the extension property. □

Uniqueness: the back-and-forth argument. Let $R = (V, E)$ and $R' = (V', E')$ be countable graphs with the extension property, and enumerate both vertex sets. We construct an increasing sequence of finite partial isomorphisms.

At a *forth* step, choose the first vertex $x \in V$ not yet in the domain. Partition the current domain into

$$U = \{u : xu \in E\}, \qquad W = \{u : xu \notin E\}.$$

The extension property in R' supplies a new vertex x' adjacent to every vertex of the image of U and none in the image of W. Map x to x'.

At the next, *back* step, reverse the roles of R and R', thereby placing the first unused vertex of V' into the image. Alternating forever produces a map whose domain is all of V, whose image is all of V', and which preserves adjacency and non-adjacency. Its union is an isomorphism $R \simeq R'$. □

6.4 The countable random graph

The model $\mathcal{G}(\mathbb{N}, p)$ is defined on vertex set $\mathbb{N}$ by including every edge independently with a fixed probability $p \in (0, 1)$.

Theorem 6.6 (Erdős–Rényi, 1963)**.** *For every fixed $p \in (0, 1)$,*

$$\mathbb{P}(G \simeq R) = 1 \qquad \text{for } G \in \mathcal{G}(\mathbb{N}, p).$$

Thus the isomorphism type of the countable random graph is independent of the particular value of $p \in (0, 1)$, with probability one.

Proof. Fix finite disjoint $U, W \subseteq \mathbb{N}$, with sizes u, w. Every vertex outside $U \cup W$ realizes the required extension pattern with the same positive probability $c = p^u q^w$. For any m distinct candidates, the probability that all fail is $(1 - c)^m$, which tends to 0. Hence the probability that no witness exists is 0.

There are only countably many ordered pairs of finite subsets (U, W) of $\mathbb{N}$. A countable union of probability-zero events still has probability zero. Therefore G has every finite extension property with probability one, and Theorem 6.5 implies $G \simeq R$. □

7 Threshold functions and the second moment

7.1 Thresholds

From now on $p = p(n)$ may tend to zero.

Definition 7.1. Let $\mathcal{P}$ be a graph property. A positive function $t = t(n)$ is a *threshold function* for $\mathcal{P}$ if

$$\mathbb{P}(G \in \mathcal{P}) \longrightarrow \begin{cases} 0, & p/t \longrightarrow 0, \\ 1, & p/t \longrightarrow \infty, \end{cases} \qquad G \in \mathcal{G}(n, p).$$

The definition identifies an asymptotic scale rather than a sharp transition window. Multiplying t by any fixed positive constant produces an equivalent threshold function.

7.2 Variance and the second-moment criterion

Definition 7.2. For a random variable X with mean $\mu = \mathbb{E}X$, its variance is

$$\mathrm{Var}(X) = \sigma^2 := \mathbb{E}\big((X-\mu)^2\big).$$

This is the second *central* moment of X.

Lemma 7.3 (Chebyshev's inequality)**.** *For every $\lambda > 0$,*

$$\mathbb{P}(|X-\mu| \geq \lambda) \leq \frac{\sigma^2}{\lambda^2}.$$

Proof. Apply Markov's inequality to the nonnegative random variable $(X-\mu)^2$. □

Lemma 7.4 (Second-moment criterion)**.** *Let $X = X_n \geq 0$, and suppose that $\mu = \mathbb{E}X > 0$ for large n. If*

$$\frac{\mathrm{Var}(X)}{(\mathbb{E}X)^2} \longrightarrow 0,$$

then $\mathbb{P}(X > 0) \to 1$.

Proof. If $X = 0$, then $|X-\mu| \geq \mu$. Chebyshev's inequality gives

$$\mathbb{P}(X=0) \leq \mathbb{P}(|X-\mu| \geq \mu) \leq \frac{\mathrm{Var}(X)}{\mu^2} \longrightarrow 0.$$

□

The first moment is effective for proving that no copy exists. The second moment is often needed to prove that a copy does exist, because it controls the dependence between overlapping potential copies.

7.3 Containing a fixed graph

For a fixed nonempty graph H, let

$$\mathcal{P}_H = \{G : H \subseteq G\}.$$

The simplest first-moment calculation uses the density of H itself.

Proposition 7.5. *Let H have $v = |H|$ vertices and $e = \|H\| \geq 1$ edges. If*

$$p \ll n^{-1/\varepsilon(H)} = n^{-v/e},$$

then almost every $G \in \mathcal{G}(n,p)$ contains no copy of H.

Proof. Let X count copies of H. There are at most n^v possible injective placements of its vertices, and each succeeds with probability p^e. Thus

$$\mathbb{E}X \leq n^v p^e = \big(pn^{v/e}\big)^e \longrightarrow 0.$$

By Markov's inequality, $\mathbb{P}(X \geq 1) \leq \mathbb{E}X \to 0$. □

If H contains a subgraph denser than H itself, that denser subgraph is the true obstruction. Define the *maximum subgraph density*

$$m(H) := \max_{\substack{J \subseteq H \\ |J| \geq 1}} \frac{\|J\|}{|J|}.$$

This is the quantity denoted $\varepsilon'(H)$ in the lecture material. A graph is called *balanced* when $m(H) = \varepsilon(H)$.

Theorem 7.6 (Erdős–Rényi; Bollobás)**.** *Let H be a fixed finite graph with at least one edge. Then*

$$t_H(n) = n^{-1/m(H)}$$

is a threshold function for the property $\mathcal{P}_H$ of containing H as a subgraph.

Proof. We prove the two threshold directions separately.

Sparse regime. Choose $J \subseteq H$ with $\|J\|/|J| = m(H)$, and write $s = |J|$, $r = \|J\|$. If $p \ll n^{-1/m(H)}$, then

$$n^s p^r = \left(pn^{1/m(H)}\right)^r \longrightarrow 0.$$

The expected number of copies of J tends to zero, so a.a.s. G contains no J, and therefore no H.

Dense regime. Suppose $p \gg n^{-1/m(H)}$. Let $v = |H|$ and $e = \|H\|$, and let X count labeled embeddings of H into $[n]$. For every injection $f : V(H) \to [n]$, let I_f indicate that all edges of the image $f(H)$ are present. Then

$$X = \sum_f I_f, \qquad \mu := \mathbb{E}X = (n)_v p^e.$$

Consider two injections f, g. If their required edge sets are disjoint, I_f and I_g are independent and their covariance is zero. Otherwise, suppose their image copies overlap in $s \geq 1$ vertices and have $r \geq 1$ required edges in common. For each fixed overlap type there are $O(n^{2v-s})$ ordered pairs (f, g), and

$$\mathbb{E}(I_f I_g) = p^{2e-r}.$$

The common part is isomorphic to a subgraph of H, so $r/s \leq m(H)$. After division by $\mu^2 = \Theta(n^{2v}p^{2e})$, the contribution of this overlap type is

$$O(n^{-s}p^{-r}) = O\left(\frac{1}{n^s p^r}\right).$$

Write $p = n^{-1/m(H)}\omega(n)$, where $\omega(n) \to \infty$. Then

$$n^s p^r = n^{s-r/m(H)}\omega(n)^r \longrightarrow \infty,$$

because $r/s \leq m(H)$. Overlaps with $r = 0$, if included in the estimate, contribute only $O(n^{-s})$. Since there are finitely many overlap types,

$$\frac{\mathrm{Var}(X)}{\mu^2} \longrightarrow 0.$$

Lemma 7.4 now gives $\mathbb{P}(X > 0) \to 1$, which is precisely $\mathbb{P}(H \subseteq G) \to 1$. □

7.4 Examples and concluding threshold landmarks

Example 7.7 (Cycles)**.** Every nonempty subgraph of C_k has density at most 1, and $\varepsilon(C_k) = 1$. Hence $m(C_k) = 1$, so

$$n^{-1}$$

is a threshold function for containing a k-cycle.

Example 7.8 (Trees)**.** Let T be a tree of order k. A subgraph on s vertices has at most $s-1$ edges, and the ratio $(s-1)/s$ increases with s. Therefore

$$m(T) = \frac{k-1}{k},$$

and the threshold for containing T is

$$n^{-k/(k-1)}.$$

Example 7.9 (Complete graphs)**.** For K_r, the densest subgraph is K_r itself and $m(K_r) = (r-1)/2$. Thus the threshold for containing K_r is $n^{-2/(r-1)}$.

The lecture concluded with two global threshold landmarks. They do not follow from Theorem 7.6, since the structures involved grow with n:

- $\log n/n$ is a threshold function for connectedness.
- For every fixed $\varepsilon > 0$, if $p \geq (1+\varepsilon)\log n/n$, then $G \in \mathcal{G}(n,p)$ is Hamiltonian a.a.s.

More precisely, the sharp Hamiltonicity transition occurs around $(\log n + \log\log n)/n$; the displayed sufficient condition lies above that window.

Further reading

The organization and notation of these notes follow the random-graph chapter of Reinhard Diestel's *Graph Theory*. The principal historical results used above are Erdős's probabilistic lower bound for Ramsey numbers (1947), his construction of graphs with simultaneously large girth and chromatic number (1959), the foundational random-graph work of Erdős and Rényi (1960–1963), and the fixed-subgraph threshold theorem developed further by Bollobás (1981).